

SMARTRIS FIRST CONFIGURATION

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1. | OBJECT CANOPEN

1.1 STANDARD OBJECT DSP402

The following table describes the standard objects of DSP402. These objects are necessary to control drive in profile velocity mode. The objects are a default values saved in e²prom.

Index	Sub Index	Name	READ / WRITE	DEFAULT EEPROM for "SMALL" RELEASE	DESCRIPTION
0x603F	0	Error Code	RO	-	This object shall provide the error code of the last error which occurred in the drive device.
0x6040	0	Control Word	R/W	-	This object shall indicate the received command controlling the PDS FSA.
0x6041	0	Status Word	RO	-	This object shall provide the status of the PDS FSA.
0x6060	0	Modes of Operation	R/W	-	This object shall indicate the requested operation mode
0x6061	0	Modes of Operation Display	RO	-	This object shall provide the actual operation mode.
0x606B	0	Velocity Demand Value	RO	-	This object shall provide the output value of the trajectory generator. The value shall be given in the user-defined velocity units.
0x606C	0	Velocity Actual Value	RO	-	This object shall provide the actual velocity value derived either from the velocity sensor or the position sensor. The value shall be given in user-defined velocity units.
0x606D	0	Velocity Window	R/W	13653 inc/sec	This object shall indicate the configured velocity window. The value shall be given in userdefined velocity units.
0x606E	0	Velocity Window Time	R/W	100 ms	This object shall indicate the configured velocity window time. The value shall be given in milliseconds.
0x606F	0	Velocity Threshold	R/W	1365 inc/s	This object shall indicate the configured velocity threshold. The value shall be given in userdefined velocity units
0x6070	0	Velocity Threshold Time	R/W	100 ms	This object shall indicate the configured velocity threshold time. The value shall be given in milliseconds.
0x607E	0	Polarity	R/W	0 (positive)	This object shall indicate if the position demand value shall be multiplied by 1 of by -1.
0x607F	0	Max Profile Velocity	R/W	4456 rpm	This object shall indicate the configured maximal allowed velocity in either direction during a profiled motion.
0x6083	0	Profile Acceleration	R/W	273066 inc/s ²	This object shall indicate the configured acceleration. The value shall be given in user-defined acceleration units.
0x6084	0	Profile Deceleration	R/W	273066 inc/s ²	This object shall indicate the configured deceleration. The value shall be given in user-defined acceleration units.
0x6096	1	Num Velocity Factor	R/W	1	This object can be used to match the velocity units to the user-defined velocity units
	2	Div Velocity Factor	R/W	1	
0x 6097	1	Num Acceleration Factor	R/W	1	This object can be used to match the acceleration units to the user-defined acceleration units.
	2	Div Acceleration Factor	R/W	1	
0x60C5	0	Max Acceleration	R/W	608393 inc/s ²	This object shall indicate the configured maximal acceleration. It is used to limit the acceleration to an acceptable value in order to prevent the motor and the moved mechanics from being destroyed. The value shall be given in user-defined acceleration physical units.
0x60C6	0	Max Deceleration	R/W	608393 inc/s ²	This object shall indicate the configured maximal deceleration. It is used to limit the acceleration to an acceptable value in order to prevent the motor and the moved mechanics from being destroyed.
0x60FD	0	Digital Inputs	RO	-	This object shall provide digital inputs.
0x60FE	0	Digital Outputs	RO	-	This object shall command the digital outputs.
0x60FF	0	Target Velocity	R/W	-	This object shall indicate the configured target velocity and shall be used as input for the trajectory generator. The value shall be given in user-defined velocity units.

1.2 MANUFACTURER OBJECT

The following table describes the manufacturer objects. These objects are customized by "Lafert Servo Drive" but these are necessary to control in profile velocity mode. There are objects with a default value saved in e²prom.

Index	Sub Index	Name	R/W	DEFAULT EEPROM	DESCRIPTION
0x2000	0	ID Node	R/W	1	The object allows the user to set the CAN IdNode of the Node
0x2001	0	CAN Baud Rate	R/W	1000	The object allows the user to set the CAN bit rate of the Node
0x2002	0	Drive Status	RO	-	This object defines exactly the PWM control motor.
0x2004	0	State Lafert Servo Drive Machine	RO	-	This object describes exactly the drive state.
0x3002	1	Motor Brake Option	R/W	1 (present)	Brake Configuration: 0: it is not present 1: it is present
	2	Motor Brake Delay	R/W	20 [10*ms]	This time is the delay between STBY Status and unlock
	3	Brake Unlock time	R/W	10 [10*ms]	Delay between STOP and RUN mode before unlock Brake. This time depends by kind of motor brake.
	4	Brake Timeout	R/W	200 [10*ms]	Only without Dynamic Brake. Max time programmed for natural Inertia deceleration. At the end of this time the brake is locked and drive will be in STBY status.
	5	Automatic/Manual Mode Configuration	R/W	0 (auto)	Configuration Brake (1 = Manual, 0 = Automatic)
	6	Motor Brake Status	RO	-	Brake Status: 0: activated (Motor is locked) 1: released (Motor is not locked)

2. | CONFIGURATION

The default value can be changed.

2.1 HOW TO CHANGE ID-NODE

Id-Node has default Value = 1. The following steps describe how to change the Id-Node.

PROCEDURE TO SET NEW ID-NODE VALUE (WRITE SDO)

(See CANopen Manual page 52)

The PLC sends SDO message ID = 0x601 (defined 0x600 + Id node)

- Data Value "command" = 0x2F
- Data Value "Index" = 0x2000
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = new Id-Node value (for Example 3)

The drive answers SDO message ID = 0x581 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x2000
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

The following picture shows the SDO message:

ID	Name	Node	Transfer data	Interpretation	Data
601	CSDO_001	Node1	03	[2000,00] Initiate Download Rq. expedited	2F 00 20 00 03 00 00 00
581	SSDO_001	Node1		[2000,00] Initiate Download Rsp	60 00 20 00 00 00 00 00

PROCEDURE TO SAVE NEW VALUE IN EEPROM (WRITE SDO)

(See CANopen Manual page 49)

IMPORTANT: during STORE procedure, drive must not be in "Operation Enabled" state or in "Quick Stop" state.

The PLC sends SDO message ID = 0x601 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x1010 (Store)
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0x73617665 (means "save" in ASCII code)

The drive answers SDO message ID = 0x581 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x1010

- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0

The following picture shows the SDO message:

ID	Name	Node	Transfer data	Interpretation	Data
601	CSDO_001	Node1	73 61 76 65	[1010,01] Initiate Download Rq. expedited "save"	23 10 10 01 73 61 76 65
581	SSDO_001	Node1		[1010,01] Initiate Download Rsp	60 10 10 01 00 00 00 00

AFTER STORED PARAMETERS PROCEED WITH

RESET ALL NODES (NMT PROTOCOL)

(See CANopen Manual page 30)

The PLC sends message ID = 0x00 (NMT protocol)

- Data Value "command" = 0x81
- Data Value "Index" = 0x00

The following picture shows the SDO message:

Dir	ID	Name	Node	Transfer data	Interpretation	Error	Data
Rx	0	NMTZeroMsg			Reset all nodes		81 00

AFTER RESET (NMT PROTOCOL)

(See CANopen Manual page 31)

The drive answers message BOOT-UP message ID = 0x703 (defined 0x700 + Id node)

- Data Value "Index" = 0x00

ID	Name	Node	Transfer data	Interpretation	Error	Data
703			Boot-up			00
703	—		Boot-up			00

THEN THE DRIVE SENDS EMERGENCY MESSAGES (EMERGENCY PROTOCOL)

(See CANopen Manual page 20)

The drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x0
- Data Value "Reg" = 0x0
- "Data" = 0

It means "ERROR RESET or NO ERROR"

And the drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x8107
- Data Value "Reg" = 0x11
- Data Value "Data" = 0

It means "CAN STATE: ACTIVE"

ID	Name	Node	Transfer data	Interpretation	Error	Data
83	EMCY_003		00 00 00 00 00 00 00 00	Error reset or no error	E	00 00 00 00 00 00 00 00
83	EMCY_003		70 81 11 00 00 00 00 00	Communication - generic	E	70 81 11 00 00 00 00 00

THEN, TO BE SURE, THE DRIVE ACCEPTED PREVIOUS ID-NODE CHANGED

PROCEDURE TO VERIFY NEW ID-NODE → READ SDO

The PLC sends SDO message ID = 0x603 (defined 0x600 + New Id node)

- Data Value "command" = 0x40
- Data Value "Index" = 0x2000
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0x00

The drive answers SDO message ID = 0x583 (defined 0x580 + New Id node)

- Data Value "command" = 0x4F
- Data Value "Index" = 0x2000
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0x3

The following picture shows the SDO messages:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003			[2000,00] Initiate Upload Rq.		40 00 20 00 00 00 00 00
583	SSDO_003	03		[2000,00] Initiate Upload Rsp. expedited		4F 00 20 00 03 00 00 00

2.2 HOW TO CHANGE BAUDRATE

Baudrate Default is 1000K. The following steps describe how to change the BaudRate.

PROCEDURE TO SET NEW BAUDRATE VALUE → WRITE SDO

(See CANopen Manual page 52)

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x2B
- Data Value "Index" = 0x2001
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = new BaudRate (for Example 500K = 0x01F4)

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x2001
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

The following picture shows the SDO messages:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		f4 01	[2001,00] Initiate Download Rq. expedited		2B 01 20 00 F4 01 00 00
583	SSDO_003			[2001,00] Initiate Download Rsp		60 01 20 00 00 00 00 00

PROCEDURE TO SAVE NEW VALUE IN EEPROM (WRITE SDO)

(See CANopen Manual page 49)

IMPORTANT: during STORE procedure, drive must not be in "Operation Enabled" state or in "Quick Stop" state.

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x1010 (Store)
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0x73617665 (means "save" in ASCII code)

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x1010
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0

The following picture shows the SDO message:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		73 61 76 65	[1010,01] Initiate Download Rq. expedite...		23 10 10 01 73 61 76 65
583	SSDO_003			[1010,01] Initiate Download Rsp		60 10 10 01 00 00 00 00

AFTER STORED PARAMETERS PROCEED WITH

RESET ALL NODES (NMT PROTOCOL)

(See CANopen Manual page 30)

The PLC sends message ID = 0x00 (NMT protocol)

- Data Value "command" = 0x81
- Data Value "Index" = 0x00

The following picture shows the SDO message:

Dir	ID	Name	Node	Transfer data	Interpretation	Error	Data
Rx	0	NMTZeroMsg			Reset all nodes		81 00

AFTER RESET (NMT PROTOCOL)

(See CANopen Manual page 31)

The drive answers message BOOT-UP message ID = 0x703 (defined 0x700 + Id node)

- Data Value "Index" = 0x00

ID	Name	Node	Transfer data	Interpretation	Error	Data
703				Boot-up		00
703	—			Boot-up		00

Then the drive sends EMERGENCY MESSAGES (EMERGENCY PROTOCOL)

(See CANopen Manual page 20)

The drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x0
- Data Value "Reg" = 0x0
- "Data" = 0

It means "ERROR RESET or NO ERROR"

And the drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x8107
- Data Value "Reg" = 0x11
- Data Value "Data" = 0

It means "CAN STATE: ACTIVE"

ID	Name	Node	Transfer data	Interpretation	Error	Data
83	EMCY_003		00 00 00 00 00 00 00 00	Error reset or no error	E	00 00 00 00 00 00 00 00
83	EMCY_003		70 81 11 00 00 00 00 00	Communication - generic	E	70 81 11 00 00 00 00 00

THEN, TO BE SURE, THE DRIVE ACCEPTED PREVIOUS ID-NODE CHANGED

PROCEDURE TO VERIFY NEW BAUDRATE → READ SDO

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x40
- Data Value "Index" = 0x2001
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0x00

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x4B
- Data Value "Index" = 0x2000
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0x01F4

The following picture shows the SDO messages:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDQ_003			[2001,00] Initiate Upload Rq.		40 01 20 00 00 00 00 00
583	SSDQ_003		f4 01	[2001,00] Initiate Upload Rsp. expedited		4B 01 20 00 F4 01 00 00

2.3 HOW TO CHANGE THE USER UNITS

Lafert Servo Drive has a default unit [inc/s] for velocity objects and [inc/s²] for acceleration objects.

If it is necessary to change the user unit (for example in [rpm] for velocity objects and [rpm/s] for acceleration objects) it has to change the factory group object. (See CANopen Manual page 40)

The velocity factory group is

$$Velocity\ Factor = \frac{Numerator}{Divisor}$$

Numerator and divisor of the Velocity Factor has to be entered separately.

The default value is [inc/s]. The numerator and the divisor are set "1" in e²prom.

To change the default unit it has to write the numerator and divisor in the object index 0x6096 and save in e²prom the new value:

$$Velocity[internal\ unit] = Velocity[user\ unit] \times \left(\frac{60}{Resolution} \right) \times \left(\frac{Numerator}{Divisor} \right)$$

Example:

The speed-set point provision is to be made in revolutions per minute (rpm).

$$Velocity[inc/sec] = Velocity[rpm] \times \left(\frac{60}{Resolution} \right) \times \left(\frac{Numerator}{Divisor} \right)$$

If the resolution of encoder is 2¹³ = 16384 then the Numerator is 16384 and the Divisor is 60

The Acceleration Factory Group has the same consideration (object index 0x6097)

PROCEDURE TO SET SET NEW FACTORY GROUP VALUES → WRITE SDO

Write **NUMERATOR** Velocity Factory Group (value = 16384):

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x6096
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 16384 = 0x4000

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x6096
- Data Value "Index" = 0x01
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

Write **DIVISOR** Velocity Factory Group (value = 60):

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x6096
- Data Value "Sub-Index" = 0x02
- Data Value "Data" = 60 = 0x3C

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x6096
- Data Value "Index" = 0x02
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

Write **NUMERATOR** Acceleration Factory Group (value = 16384):

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x6097
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 16384 = 0x4000

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x6097
- Data Value "Index" = 0x01
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

Write **DIVISOR** Acceleration Factory Group (value = 60):

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x6097
- Data Value "Sub-Index" = 0x02
- Data Value "Data" = 60 = 0x3C

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x6097
- Data Value "Index" = 0x02
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

The following picture shows the SDO messages:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		00 40 00 00	[6096,01] Initiate Download Rq. expedited ".@.."		23 96 60 01 00 40 00 00
583	SSDO_003			[6096,01] Initiate Download Rsp		60 96 60 01 00 00 00 00
603	CSDO_003	3c 00 00 00		[6096,02] Initiate Download Rq. expedited "<..."		23 96 60 02 3C 00 00 00
583	SSDO_003			[6096,02] Initiate Download Rsp		60 96 60 02 00 00 00 00
603	CSDO_003		00 40 00 00	[6097,01] Initiate Download Rq. expedited ".@.."		23 97 60 01 00 40 00 00
583	SSDO_003			[6097,01] Initiate Download Rsp		60 97 60 01 00 00 00 00
603	CSDO_003	3c 00 00 00		[6097,02] Initiate Download Rq. expedited "<..."		23 97 60 02 3C 00 00 00
583	SSDO_003			[6097,02] Initiate Download Rsp		60 97 60 02 00 00 00 00

Now the user units are ready to save in Eeprom, but they aren't available. It MUST STORE and RESET.

PROCEDURE TO SAVE NEW VALUE IN EEPROM (WRITE SDO)

(See CANopen Manual page 49)

IMPORTANT: during STORE procedure, drive must not be in "Operation Enabled" state or in "Quick Stop" state.

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x1010 (Store)
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0x73617665 (means "save" in ASCII code)

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x1010
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0

The following picture shows the SDO message:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		73 61 76 65	[1010,01] Initiate Download Rq. expedite...		23 10 10 01 73 61 76 65
583	SSDO_003			[1010,01] Initiate Download Rsp		60 10 10 01 00 00 00 00

AFTER STORED PARAMETERS PROCEED WITH

RESET ALL NODES (NMT PROTOCOL)

(See CANopen Manual page 30)

The PLC sends message ID = 0x00 (NMT protocol)

- Data Value "command" = 0x81
- Data Value "Index" = 0x00

The following picture shows the SDO message:

Dir	ID	Name	Node	Transfer data	Interpretation	Error	Data
Rx	0	NMTZeroMsg			Reset all nodes		81 00

AFTER RESET (NMT PROTOCOL)

(See CANopen Manual page 31)

The drive answers message BOOT-UP message ID = 0x703 (defined 0x700 + Id node)

- Data Value "Index" = 0x00

ID	Name	Node	Transfer data	Interpretation	Error	Data
703			Boot-up			00
703	—		Boot-up			00

Then the drive sends EMERGENCY MESSAGES (EMERGENCY PROTOCOL)

(See CANopen Manual page 20)

The drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x0
- Data Value "Reg" = 0x0
- "Data" = 0

It means "ERROR RESET or NO ERROR"

And the drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x8107
- Data Value "Reg" = 0x11
- Data Value "Data" = 0

It means "CAN STATE: ACTIVE"

ID	Name	Node	Transfer data	Interpretation	Error	Data
83	EMCY_003		00 00 00 00 00 00 00 00	Error reset or no error	E	00 00 00 00 00 00 00 00
83	EMCY_003		70 81 11 00 00 00 00 00	Communication - generic	E	70 81 11 00 00 00 00 00

2.4 OBJECT WITH DIFFERENT DEFAULT

If the factory group changed than it is mandatory to change in [user unit] all values related to velocity and acceleration/deceleration of velocity profile mode and save in e²prom.

After reset or poweron the drive initializes the object from eeprom.

It has to change the following objects:

Index	Sub Index	Name Object	Default Value [internal unit]	New Value [user unit]
0x606D	0	Velocity Window	13653 inc/sec	50 rpm
0x606F	0	Velocity Threshold	1365 inc/s	5 rpm
0x6083	0	Profile Acceleration	273066 inc/s ²	1000 rpm/s
0x6084	0	Profile Deceleration	273066 inc/s ²	1000 rpm/s
0x60C5	0	Max Acceleration	608393 inc/s ²	2228 rpm/s
0x60C6	0	Max Deceleration	608393 inc/s ²	2228 rpm/s

PROCEDURE TO SET NEW VALUES IN USER UNIT → WRITE SDO

For example save Max Acceleration: new value 2228 [rpm/s]

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x60C5
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 2228 = 0x08B4

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x60C5
- Data Value "Index" = 0x00
- Data Value "Sub-Index" = 0x00
- Data Value "Data" = 0

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		b4 08 00 00	[60c5,00] Initiate Download Rq. expedited		23 C5 60 00 B4 08 00 00
583	SSDO_003			[60c5,00] Initiate Download Rsp		60 C5 60 00 00 00 00 00

It has to write all objects with new values in [user unit]. It has to use the same procedure.

PROCEDURE TO SAVE NEW VALUE IN EEPROM (WRITE SDO)

(See CANopen Manual page 49)

IMPORTANT: during STORE procedure, drive must not be in "Operation Enabled" state or in "Quick Stop" state.

The PLC sends SDO message ID = 0x603 (defined 0x600 + Id node)

- Data Value "command" = 0x23
- Data Value "Index" = 0x1010 (Store)
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0x73617665 (means "save" in ASCII code)

The drive answers SDO message ID = 0x583 (defined 0x580 + Id node)

- Data Value "command" = 0x60
- Data Value "Index" = 0x1010
- Data Value "Sub-Index" = 0x01
- Data Value "Data" = 0

The following picture shows the SDO message:

ID	Name	Node	Transfer data	Interpretation	Error	Data
603	CSDO_003		73 61 76 65	[1010,01] Initiate Download Rq. expedite...		23 10 10 01 73 61 76 65
583	SSDO_003			[1010,01] Initiate Download Rsp		60 10 10 01 00 00 00 00

AFTER STORED PARAMETERS PROCEED WITH

RESET ALL NODES (NMT PROTOCOL)

(See CANopen Manual page 30)

The PLC sends message ID = 0x00 (NMT protocol)

- Data Value "command" = 0x81
- Data Value "Index" = 0x00

The following picture shows the SDO message:

Dir	ID	Name	Node	Transfer data	Interpretation	Error	Data
Rx	0	NMTZeroMsg			Reset all nodes		81 00

AFTER RESET (NMT PROTOCOL)

(See CANopen Manual page 31)

The drive answers message BOOT-UP message ID = 0x703 (defined 0x700 + Id node)

- Data Value "Index" = 0x00

ID	Name	Node	Transfer data	Interpretation	Error	Data
703			Boot-up			00
703	—		Boot-up			00

Then the drive sends EMERGENCY MESSAGES (EMERGENCY PROTOCOL)

(See CANopen Manual page 20)

The drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x0
- Data Value "Reg" = 0x0
- "Data" = 0

It means "ERROR RESET or NO ERROR"

And the drive sends message ID = 0x83 (defined 0x80 + Id node)

- Data Value "Error Code" = 0x8107
- Data Value "Reg" = 0x11
- Data Value "Data" = 0

It means "CAN STATE: ACTIVE"

ID	Name	Node	Transfer data	Interpretation	Error	Data
83	EMCY_003		00 00 00 00 00 00 00 00	Error reset or no error	E	00 00 00 00 00 00 00 00
83	EMCY_003		70 81 11 00 00 00 00 00	Communication - generic	E	70 81 11 00 00 00 00 00

REVISION HISTORY:

Rel.	Date	Description
1.0	02/09/2020	First Release